Lavannya Pradeep Rathod

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SUMMARY

Recent graduate skilled in Python and SQL with hands-on experience in data science, machine learning algorithms, data modeling, and AI-based applications. Proficient in predictive modeling, predictive analytics, data preprocessing, feature engineering, statistical analysis, and exploratory data analysis (EDA). Experienced in building data-driven solutions and data visualizations using Pandas, Scikit-learn, and Power BI.

SKILLS

- **Programming:** Python, SQL, HTML
- **Data Science & Analysis:** Pandas, NumPy, Data Cleaning, Feature Engineering, Data Modeling, Statistical Analysis, Exploratory Data Analysis (EDA)
- **Machine Learning:** Scikit-learn, Machine Learning Algorithms, Logistic Regression, Random Forest, Model Evaluation, Classification, Predictive Analytics
- **Data Visualization:** Power BI, Tableau, Matplotlib, Microsoft Excel
- **Deep Learning:** LSTM, Time-Series Analysis, Neural Networks
- **Web & Deployment:** Streamlit, REST APIs, Model Deployment
- **Tools:** Git, MySQL, Jupyter Notebook, Visual Studio Code

PROJECTS

1. Heart Disease Detection System

Tools: Python, Pandas, Scikit-learn, Machine Learning, Streamlit

- Developed a machine learning model using Logistic Regression and Random Forest on a dataset with 300+ records and 13+ clinical features
- Executed data preprocessing, feature engineering, and exploratory data analysis (EDA) to enhance model performance
- Analyzed model effectiveness, achieving approximately 85% accuracy with strong precision and recall metrics
- Deployed a Streamlit-based interface enabling real-time predictions derived from user-provided health parameters
- Applied machine learning algorithms and predictive analytics for classification and forecasting tasks

2. AI Blog Generator

Tools: Python, Streamlit, OpenAI API, HTML/CSS

- Engineered an AI-powered web application using Streamlit and OpenAI APIs capable of generating complete blog articles of 500+ words in under 5 seconds
- Constructed backend logic in Python for prompt processing and dynamic content generation, and enabled export functionality supporting 3 formats including TXT, HTML, and Markdown
- Designed an interactive UI supporting 5+ configurable prompt parameters for seamless, real-time content generation

3. E-commerce Customer Behavior Analysis

Tools: Python, Pandas, NumPy, Matplotlib, Power BI

- Performed exploratory data analysis (EDA) and statistical analysis on an e-commerce dataset with 1000+ transaction records
- Standardized data cleaning and preprocessing techniques across 10+ features for structured evaluation
- Produced 5+ data visualization dashboards with Power BI and Matplotlib to highlight key business insights
- Derived actionable insights that informed business decision-making and improved customer engagement strategies

4. Automated Data Analytics Platform

Tools: Python, Pandas, NumPy, Plotly, NLP

- Built a Python-based data science platform processing Excel/CSV datasets with 500+ records and 15+ input features
- Streamlined data cleaning, statistical analysis, and data visualization workflows across 5+ metrics, integrating NLP for text-based data analysis
- Generated interactive reports with 4+ visual insights, cutting manual reporting effort and supporting faster data-driven decision-making

EXPERIENCE

Computer Literacy Volunteer: Government College

- Taught basic computer skills including MS Office, internet usage, and typing to 30+ students across 15+ sessions
- Mentored learners in fundamental programming concepts and delivered hands-on guidance during 10+ lab sessions
- Resolved 20+ technical issues related to software and system usage, ensuring smooth lab operations
- Strengthened digital literacy and confidence among 30+ students through 10+ hands-on practical learning activities

EDUCATION

The Oxford College of Engineering, Bangalore

Bachelor of Engineering in Electronics and Communication Engineering — Dec 2021 – May 2025

CGPA: 7.9/10

CERTIFICATIONS

- SQL for Beginners – MySQL and Database Design (Udemy)
- Microsoft Excel – From Beginner to Expert
- Microsoft Power BI – Advanced Data Transformation
- SQL (Advanced) – HackerRank